- Please note that some errors are extremely subjective and varies over the correct progress of the answers given. The marks are adjusted accordingly.

- Question-1 :

  - Correct answer : $\frac{e^2}{e^2+4}$ [Full marks]

  - "Correctly written probability with Bayes theorem but mistaken at writing the answer at the last line" [0.5 (major mistake)
    1 (minor mistake/calculation error)]

  - Wrong approach is awarded 0.

Question-2:

- Joint distribution :

$$\pi(\underset{\sim}{x}) \propto \frac{1}{|2\pi\Sigma|^{T/2}(1-\rho^2)^{\frac{(T-1)p}{2}}} \times$$

$$\exp -\frac{1}{2}\left\{\underset{\sim}{x}_1^T \Sigma^{-1} \underset{\sim}{x}_1 + \frac{1}{1-\rho^2}\sum_{t=2}^{T}(\underset{\sim}{x}_t - \rho\underset{\sim}{x}_{t-1})^T \Sigma^{-1}(\underset{\sim}{x}_t - \rho\underset{\sim}{x}_{t-1})\right\}$$

- Correct answer (1.5).
- Subjective partial marks are given for "red"

multiplier (0.5 or 1 accordingly).

- Wrong approach / answer $\boxed{0}$

(B) Conditional distribution:

$$\underset{\sim}{x}_t \mid \underset{\sim}{x}_1, \dots, \underset{\sim}{x}_{t-1}, \underset{\sim}{x}_{t+1}, \dots, \underset{\sim}{x}_T$$

$$\sim \text{MVN}\left(\frac{\rho}{1+\rho^2}\left(\underset{\sim}{x}_{t-1} + \underset{\sim}{x}_{t-1}\right), \frac{1-\rho^2}{1+\rho^2}\Sigma\right)$$

- Correct answer $\boxed{2.5}$
- Silly mistake at the multiplier [1.5]
- Wrong answer [0]

3.

- Posterior distribution (1 mark)

$$\lambda \mid x_1, x_2, \dots, x_T \sim G\left(T+a, b+2x_1+3\sum_{t=2}^{T} x_{t-1}x_t\right)$$

- Posterior mean

$$\mathbb{E}\left[\lambda \mid x_1, x_2 \dots x_T\right] = \frac{T+a}{b+2x_1+3\sum_{t=2}^{T} x_{t-1}x_t} \quad (0.5)$$

- (No partial marks).

4.

$\underset{\sim}{\theta} \mid \underset{\sim}{Y}_1 = \underset{\sim}{y}_1, \underset{\sim}{Y}_2 = \underset{\sim}{y}_2 \sim \text{Dir}(10, 8, 5)$. (1 mark).

Part (b)

$$\boxed{\frac{\Gamma(11)}{\Gamma(5)\Gamma(5)\Gamma(3)}} \times \frac{\Gamma(23)}{\Gamma(10)\Gamma(8)\Gamma(5)} \times \frac{\Gamma(14)\Gamma(12)\Gamma(17)}{\Gamma(33)}$$

(2 marks)

$\Downarrow$

Same as $= \dfrac{10!}{4!\,4!\,2!}$

[Partial marks if some one make "last line" error in "red" part and the rest of expression correct]

Part (c)

Correct answer

$$= \boxed{\frac{\Gamma(11)}{\Gamma(10)\Gamma(2)}} \times \frac{\Gamma(23)}{\Gamma(10)\Gamma(8)\Gamma(5)} \times \frac{\Gamma(14)\Gamma(12)\Gamma(17)}{\Gamma(33)}.$$

$\Downarrow$

$= 10 \times \cdots$

(2 marks).

[Partial marks if some one make "last line" error in "red" part and the rest of expression correct]

5. [Prior mean and prior variance, not posterior mean and posterior variance].

$\mathbb{E}_\pi(\theta) = \frac{1}{2}$ (1 mark)

$\mathrm{Var}_\pi(\theta) = \frac{1}{15}$ (1.5 marks)

[No partial marks for either].

6.

- Jeffreys prior: $\pi(\beta) \propto \frac{1}{\beta}, \beta > 0$ [1.5]

  [With correct derivation].

- Posterior distribution: $G\left(n\alpha, \frac{1}{\sum \frac{1}{x_i}}\right)$ (1.5).

  [Partial marks if correctly written density function but wrong identification: such as $IG\left(n\alpha, \frac{1}{\sum \frac{1}{x_i}}\right)$]

- Derivation of correct $\hat{\beta}_{MAP} = \frac{n\alpha - 1}{\sum_{i=1}^{n} \frac{1}{x_i}}$ [1.5]

  [No partial marks].

- Unbiasedness $\mathbb{E}[\hat{\beta}_{MAP}] = \beta$ [1.5].

  [No partial marks].

- Correct $\text{Var}[\hat{\beta}_{MAP}] = \dfrac{\beta^2}{n\alpha - 2}$ [1].
- Showing $\text{Var}[\hat{\beta}_{MAP}] \longrightarrow 0$ as $n \longrightarrow \infty$ [0.5]

[No partial marks].

[⊛ Note that if one does mistake from any stage of the answer, (say wrong density function) NO MARKS IS AWARDED CONSEQUENTLY].

7.

Jeffrey's prior of θ: $\pi(\theta) \propto \dfrac{1}{\sqrt{\theta(1-\theta)}}$

$= \beta\left(\frac{1}{2}, \frac{1}{2}\right)$ (1 mark)

[No partial marks].

- $\phi = \theta^3$ (say), correct $\pi(\phi) \propto \phi^{-5/6}\left(1 - \phi^{1/3}\right)^{-1/2}$

(1 mark).

- Commenting "deriving correctly $\pi(\phi)$" on invariance (0.5 marks).

[No partial marks].